

Tien-Ning(Sky) Lee

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EDUCATION

University of California, San Diego (UCSD) Sep. 2025 - Expected Jun. 2027
M.S. in Electrical and Computer Engineering (Machine Learning & Data Science Track) California, United States

National Yang Ming Chiao Tung University (NYCU) Sep. 2020 - Jun. 2024
B.S. in Information Management and Finance (Program of Information Management) Hsinchu, Taiwan

EXPERIENCE

Classmethod Jul. 2026 - Sep. 2026
Physical AI Engineer Intern Tokyo, Japan

- Developed Physical AI applications for **Unitree G1 and Go2 robots**, including a visual inspection agent that converts live camera input into object detections, risk assessments, and operator alerts.
- Integrated **Unitree SDK2**, **CycloneDDS**, and a persistent GPU inference service for continuous frame ingestion, structured outputs, JSON validation, and automated inspection reports.
- Evaluated vision-language models and redesigned the real-time pipeline using **Grounding DINO**, **ROI geometry**, and deterministic risk policies, achieving **sub-second inference** while improving spatial grounding, reproducibility, and explainability.
- Built a real-time monitoring dashboard with detection overlays, route-occupancy analysis, risk explanations, inference telemetry, process controls, and audio warnings.

Artificial Intelligence Lab, National Taiwan University Jun. 2024 - Dec. 2024
Project Research Assistant | Advisor: **Prof. Tsung-Nan Lin (IEEE Fellow)** Taipei, Taiwan

- Co-developed **trustworthy AI recommendation systems** with PhD students for **BankTaiwan Life Insurance**, focusing on minimizing default risk through **anti-recommendation algorithms**.
- Processed large-scale real-world insurance data and **engineered robust features** to enhance model generalization.
- Implemented and evaluated collaborative filtering and **anomaly detection** models using **PyTorch**, optimizing for both ranking accuracy and risk mitigation.
- Collaborated with stakeholders to **officially deploy** the model after successful enterprise testing.

National Institute of Information and Communications Technology (NICT) Jul. 2023 - Sep. 2023
AI Research Intern Tokyo, Japan

- Conducted research to enhance the resilience of **model-based network intrusion detection systems (NIDS)** for **IoT devices** by investigating advanced **adversarial attacks** to develop robust defense strategies.
- Engineered and trained diverse NIDS models—including **Logistic Regression**, **kNN**, **Random Forest**, **Autoencoder**, **1D-CNN**, and **2D-CNN**—using **scikit-learn** and **TensorFlow** on **real-world TON_IoT datasets**.
- Developed a **SHAP**-guided feature selection algorithm that **reduced input features by 65%** while maintaining model performance and interpretability.
- Investigated adversarial training using **SHAP**-informed attacks, providing key insights that supported NICT researchers' ongoing studies on adversarial training and defense strategies.

PROJECTS

Amazon-SASRec-Engine | PyTorch, FAISS, FastAPI, Streamlit | Full-Stack Deep Recommender System

- Developed a **full-stack sequential recommendation system** using the **SASRec** (Transformer) model to predict user purchase intent based on interaction history.
- Implemented **FAISS** to achieve **sub-millisecond** inference latency for **real-time** recommendations.
- Designed a decoupled architecture with **Streamlit** (frontend) and **FastAPI** (backend), featuring an oversampling strategy to filter duplicates and ensure consistent top-k results.
- Integrated a real-time web scraper with **BeautifulSoup** as a fallback mechanism to dynamically fetch missing product metadata from Amazon, resolving cold-start display issues.

SKILLS

Languages: Mandarin Chinese (native), English (fluent), Japanese (business level, JLPT N2)

Programming Languages: Python, C++, JavaScript, Java, R, SQL

Libraries/Frameworks/Tools: scikit-learn, TensorFlow, PyTorch, React, FastAPI, Alembic, Git, AWS, Render, Docker, ROS2, pytest